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BRIEF NOTES

The Specificity of the cAMP Receptor Mediating Activation of Adenylate Cyclase in *Dictyostelium discoideum*

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In *Dictyostelium discoideum* amoebae, binding of cyclic AMP (cAMP) to surface receptors elicits numerous responses including chemotaxis, cyclic GMP (cGMP) accumulation, and activation of adenylate cyclase. The specificity of the surface cAMP receptor which mediates activation of adenylate cyclase and cAMP secretion was determined by testing the relative effectiveness of a series of 10 cAMP analogs. Each of the 10 analogs elicited cAMP secretion, chemotaxis, and cGMP accumulation in the same dose range. The order of potency for eliciting these responses (cAMP > 2′-H-cAMP > N^1-O-cAMP > cAMPS(Sp) > 6-Cl-cAMP > cAMPN$(CH_3)_2$(Sp) > 3′-NH-cAMP > 8-Br-cAMP > cAMPS(Rp) > cAMPN$(CH_3)_2$(Rp)) matches that for binding to the major cell surface cAMP binding sites and differs from that of the cell surface phosphodiesterase and the major intracellular cAMP binding protein. © 1986 Academic Press, Inc.

INTRODUCTION

In *Dictyostelium* starvation initiates development, whereby 10^6 amoebae aggregate into a multicellular structure which differentiates into stalks and spores (Devreotes, 1982; Gerisch, 1982; Van Haastert and Konijn, 1982). Spontaneously synthesized by cells in aggregation centers, cyclic AMP (cAMP) diffuses to adjacent cells, binds to surface receptors, and activates adenylate cyclase, sustaining the signal relay. The resulting waves provide gradients for chemotaxis. Evidence indicates that cAMP regulates gene expression (Darmon *et al.*, 1977; Sampson *et al.*, 1978; Chisholm *et al.*, 1984), morphogenesis (Sussman, 1982) and differentiation (Gross *et al.*, 1981).

Other responses concurrent with signaling and chemotaxis include a rise in intracellular cyclic GMP (cGMP) (Mato *et al.*, 1977a,b), a decrease in extracellular pH (Malchow *et al.*, 1978), an increase in cellular Ca^{2+} and decrease in cellular K^+ (Wick *et al.*, 1978; Bumann *et al.*, 1984), phosphorylation of myosin and the major surface cAMP receptor (Berlot *et al.*, 1985, Lubs-Haukeness and Klein, 1985), and an increase in vesicles near the cell surface (Maeda and Gerisch, 1977). These responses are all initiated by binding of cAMP to the cell surface, yet it is not known whether one class of cAMP surface receptors or different subclasses of receptors trigger each response.

The specificity of the receptors linked to activation of adenylate cyclase was determined by testing cAMP and 10 analogs for effectiveness in eliciting cAMP secretion (signaling). This data was compared to analog data for cGMP accumulation, chemotaxis, binding to surface receptors (Mato *et al.*, 1977a; Mato and Konijn, 1977; Van Haastert *et al.*, 1982; Van Haastert, 1983; Van Haastert and Kien, 1983), phosphodiesterase (Van Haastert *et al.*, 1983), and binding to the major cytoplasmic binding protein (De Wit *et al.*, 1982).

MATERIALS AND METHODS

Measurement of cAMP signaling responses. Dictyostelium discoideum strain NC-4 cells were grown and [^{3}H]cAMP secretion monitored as previously described (Devreotes *et al.*, 1979; Devreotes and Steck, 1980). Cells were developed at 7°C for 16 hr or 22°C for 3-5 hr in the dark.

Chemicals. Cyclic AMP and analogs **c**,[1] **f**, and **i** were purchased from Sigma Company (St. Louis, Mo.). The synthesis of **d**, **h**, and **b** has been previously described (Van Haastert, 1983; Van Haastert and Kien, 1983; Jastorff and Friest, 1974). Analogs **e**, **g**, **j**, **k** were synthesized in Dr. Jastorff's laboratory. [^{3}H]adenosine was from ICN (Irvine, Calif.).

Data analysis. Total [^{3}H]cAMP secreted determined at each concentration was plotted versus log [stimulus].

[1] Abbreviations used: (**a**), cAMP, adenosine 3′,5′-phosphate; (**b**), 5′-NH-cAMP, 5′-deoxy 5′-amino-adenosine 3′,5′-phosphate; (**c**) 2′-H-cAMP, 2′deoxyadenosine 3′,5′-phosphate; (**d**), N^1-O-cAMP, adenosine N^1-oxide 3′,5′-phosphate; (**e**), cAMPS(Sp), adenosine 3′,5′-phosphorothioate, Sp-stereoisomer; (**f**), 6-Cl-cPMP, 6-chloropurine riboside 3′,5′-phosphate; (**g**), cAMPN$(CH_3)_2$(Sp), adenosine 3′,5′-phosphodimethylamino-amidate, Sp; (**h**), 3′-NH-cAMP, 3′-deoxy-3′amino-adenosine 3′,5′-phosphate; (**i**), 8-Br-cAMP, 8-bromo-adenosine 3′,5′-phosphate; (**j**), cAMPS(Rp), adenosine 3′,5′-phosphorothioate, Rp; (**k**), cAMPN$(CH_3)_2$(Rp), adenosine 3′,5′-phosphodimethyl-amino-amidate, Rp.

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Responses were normalized to the saturated response except when saturation was not achieved when responses were normalized to that of a 100 n*M* cAMP control stimulus. The K_{50} is the concentration of cAMP or analog that elicits 50% of the maximum response. To facilitate the comparison of signaling data with that reported for other responses, data was converted to *R*. *R* = log (concentration of analog eliciting half-maximal response/ concentration of cAMP eliciting half-maximal response).

RESULTS

Cells labeled with [^{3}H]adenosine and developed to aggregation competence were placed on filters, and stimulated with cAMP or a cAMP analog. The cells respond by activating adenylate cyclase, the newly synthesized cAMP is rapidly secreted, then the secretion rate declines to prestimulus levels (Devreotes *et al.*, 1979) as the cells adapt to the stimulus. The total amount of [^{3}H]cAMP secreted (the cAMP signaling response) depends on the cAMP stimulus concentration, saturating at 100 n*M* cAMP (>10 μM decreases the maximum). The effectiveness of cAMP and 10 analogs (listed in Table 1) in eliciting cAMP secretion was determined. Dose-response data for these analogs are shown in Fig. 1. Modification of the cAMP molecule at different positions increased the dose required to elicit a half maximal response by varying degrees. For example, the 2′-deoxy cAMP analog (**c**) was 20-fold less effective than cAMP. Substitution of Cl for H at the 6 position (**f**) caused a 500-fold reduction in effectiveness. The concentrations required to produce 50% of the maximal response (K_{50}) for the analogs are listed in Table 1.

Modifications change the effectiveness in eliciting a signaling response, but the shape of the dose-response curve is similar for each analog and all responses appear to saturate (where sufficiently high concentrations of analog could be tested). The maximal response achieved was similar to that produced by a saturating dose of cAMP. Furthermore, at appropriate doses, responses of equivalent magnitude had identical kinetics, regardless of the analog used to stimulate.

R values (*R* = log of the concentration of analog eliciting half-maximal response divided by the concentration of cAMP which elicited a half-maximal response) calculated for signaling are presented in Table 1 together with the *R* values calculated from previously reported data for the rise in intracellular cGMP, chemotaxis, surface receptor binding antagonism (summarized in Van Haastert, 1983; Van Haastert and Kien, 1983), substrate specificity of the cAMP phosphodiesterase (Van Haastert *et al.*, 1983), and cytoplasmic cAMP binding activity (De Wit *et al.*, 1982). A graphical comparison of the signaling *R* to the other *R* values is shown in Fig. 2. Since cAMPS(Rp) and cAMPN$(CH_3)_2$(Rp) elicited only small signaling responses but not the chemotactic or cGMP response, the data for these two analogs were omitted from this graphical comparison. From Fig. 2, it is evident that there is a high degree of correlation between the potency of an analog in eliciting a cAMP signaling response and each of the other three responses. Linear regression analysis yields lines with slopes and intercepts of 0.80 and 0.060 for signaling compared with binding, 0.90 and −0.026 compared with chemotaxis, and 0.73 and 0.182 compared with cGMP accumulation. The correlation coefficients for these lines are r = 0.97, r = 0.86, and r = 0.99, respectively. This data shows that with the exception of the two Rp analogs, the analogs elicit the various responses with approximately the same order of effectiveness and at the same concentrations.

There is little correlation between the *R* values for signaling compared to surface phosphodiesterase substrate activity or binding to the intracellular cAMP

TABLE 1
COMPARISON OF *R* VALUES FOR SIGNALING AND OTHER RESPONSES

Analog	$K_{50}(M)$	Signaling	Binding	Chemotaxis	*R* values cGMP	PDE	Intracellular receptor
a cAMP	4.4×10^{-9}	0	0	0	0	0	0
b 5′-NH-cAMP	1.3×10^{-8}	0.47	0.67	0.52	0.71	−1.15	3.35
c 2′-H-cAMP	8.5×10^{-8}	1.29	0.88	1.12	1.08	0.65	4.21
d N′-O-cAMP	4.2×10^{-7}	1.98	1.61	1.43	1.79	0.45	0.90
e cAMPS(Sp)	1.5×10^{-6}	2.46	1.88	1.95	1.88	2.34	0.86
f 6-Cl-cPMP	2.4×10^{-6}	2.74	2.59	3.00	—	0.40	0.34
g cAMPN$(CH_3)_2$(Sp)	4.4×10^{-6}	3.00	2.09	1.99	2.50	1.36	3.35
h 3′-NH-cAMP	6.6×10^{-6}	3.18	2.69	4.12	2.39	>3.40	2.49
i 8-Br-cAMP	6.8×10^{-6}	3.19	2.73	2.03	2.45	0.81	−0.50
j cAMPS(Rp)	2.0×10^{-5}	3.67	1.90	—	>4.08	>3.40	2.30
k cAMPN$(CH_3)_2$(Rp)	5.4×10^{-5}	4.09	2.32	—	>4.08	2.65	5.22

Note. K_{50} is the concentration of analog which produced a half-maximal response for signaling. The *R* value is log [analog K_{50}/cAMP K_{50}]. Data for chemotaxis and cGMP accumulation and surface cAMP binding were obtained from Van Haastert (1983) and Van Haastert and Kien (1983), data for PDE were from (van Haastert *et al.*, 1983), and data for intracellular binding were from (De Wit *et al.*, 1982). PDE—phosphodiesterase.

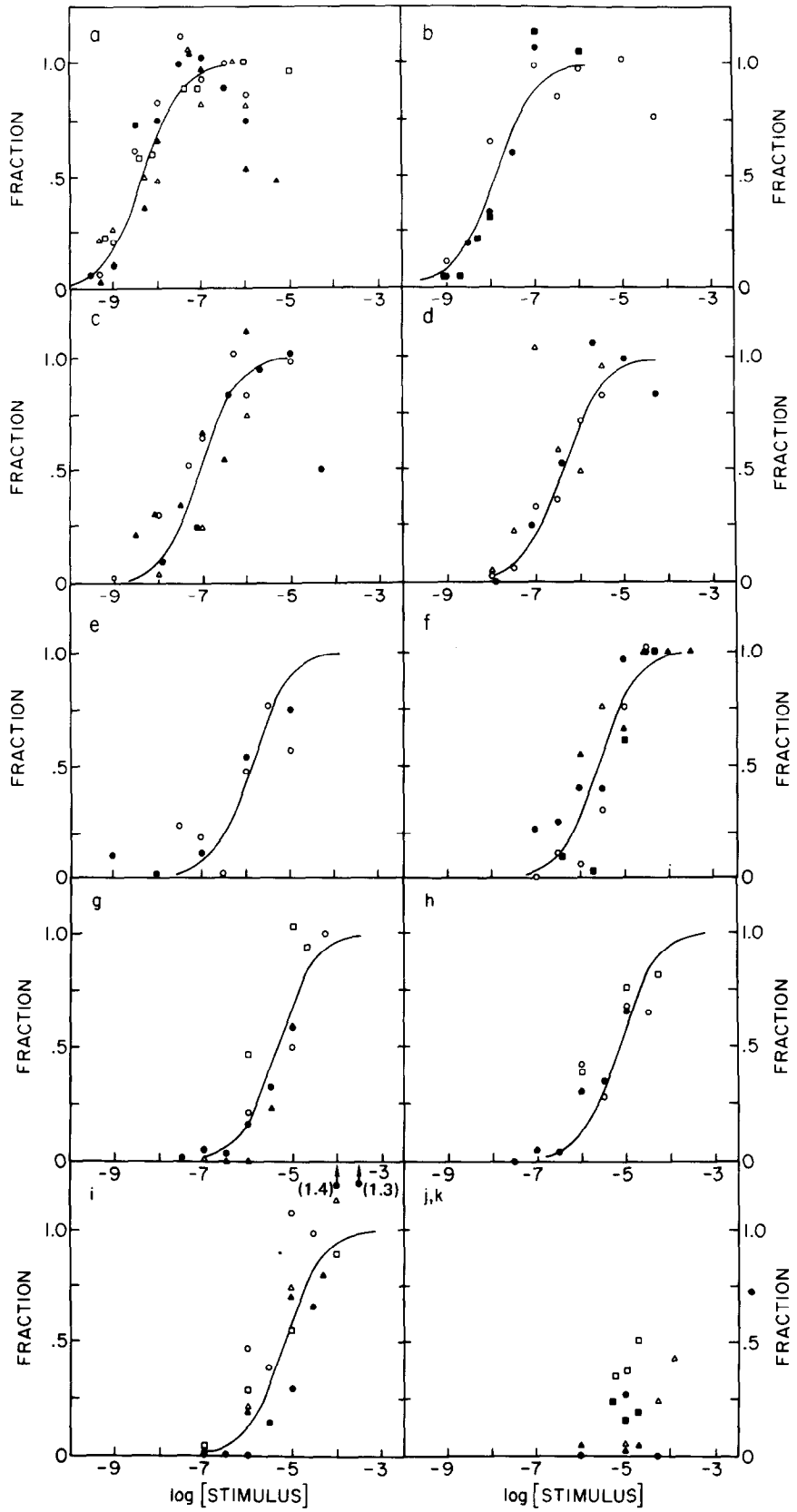

FIG. 1. Dose-response curves for cAMP and cAMP analogs. For panels (**a**)-(**i**), the stimuli were: (**a**) cAMP: (**b**) 5′-NH-cAMP, (**c**) 2′-H-cAMP, (**d**) N^1-O-cAMP (**e**) cAMPS(Sp), (**f**) 6-Cl-cPMP, (**g**) cAMPN$(CH_3)_2$(Sp), (**h**) 3′-NH-cAMP, (**i**) 8-Br-cAMP. Each data point was obtained by integrating secretion profiles for cAMP signaling and plotting vs log [stimulus]. Different symbols indicate separate experiments. Panel (**j, k**), stimuli were cAMPN$(CH_3)_2$(Rp) (●, ▲, ■), cAMPs(Rp) (△, □).

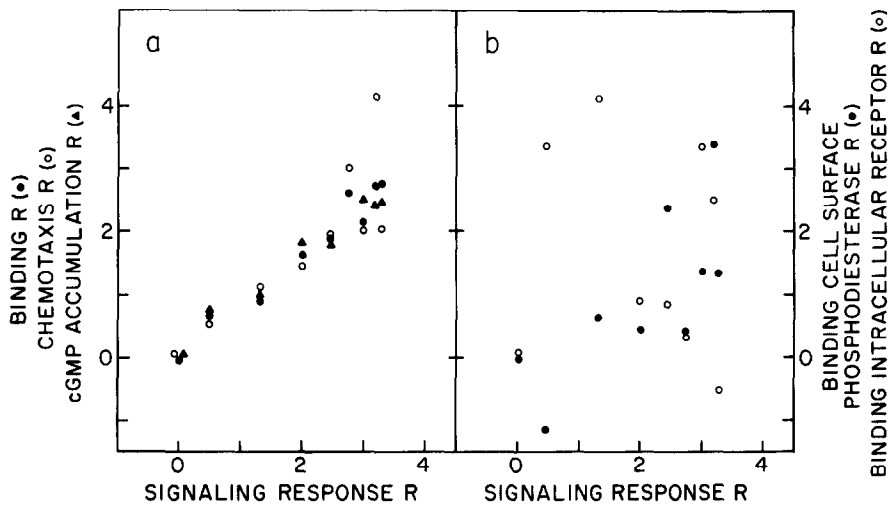

FIG. 2. Comparison of cAMP analog-elicited responses. (a) Comparison of signaling response R with binding R (●) (slope = 0.80, intercept = 0.060, r = 0.97), chemotaxis R(○) (slope = 0.90, intercept = −0.026, r = 0.86), and cGMP accumulation R(▲) (slope = 0.73, intercept = 0.182, r = 0.99). The slope and intercept for the combined data are 0.81 and 0.071, r = 0.92. (b) Comparison of signaling response R with cell surface phosphodiesterase R(●) (slope = 0.48, intercept = 0.16, r = 0.52), and binding intracellular receptor R(○) (slope = −0.20, intercept = 2.0, r = −0.14). R is defined under Methods.

binding protein shown in Fig. 2b. The slopes, intercepts, and correlation coefficients are 0.48, −0.16 and 0.52 for the comparison of signaling versus phosphodiesterase, and −0.20, 2.0, and −0.14 for signaling versus the intracellular binding protein. Chemotaxis, cGMP accumulation and the signaling response are mediated by a similar receptor which has very different specificity than either the cAMP phosphodiesterase or the intracellular cAMP binding protein.

DISCUSSION

Previous reports indicate that cGMP accumulation and chemotaxis are mediated by the same receptor (Mato *et al.*, 1977a,b; Van Haastert, 1983; Van Haastert and Kien, 1983). We have determined the specificity of the receptor that mediates the cAMP signaling response. The comparison of this data with that determined by other groups for binding antagonism, chemotaxis, and cGMP accumulation shows that the specificity of the receptor mediating each process is nearly identical. A minor discrepancy was found with the Rp analogs, cAMPS(Rp) and cAMPN$(CH_3)_2$(Rp) which reportedly elicit no chemotactic or cGMP response (Van Haastert, 1983; Van Haastert and Kien, 1983). These Rp analogs elicited small signaling responses (Fig. 1j, k) at high concentrations. Contamination of the Rp isomer with the Sp isomer or cAMP could account for this discrepancy. This seems unlikely since we used the same initial preparations as used by Van Haastert (1983), however, we cannot rule out that in subsequent handling our preparations were slightly degraded to cAMP. Alternatively, detection of very weak chemotactic and cGMP responses may be difficult.

The dose-response curves show that signaling responses saturate at high concentrations and the magnitude of the maximal response is the same as that obtained with cAMP. The signaling response, at least, appears to depend on the fractional receptor occupancy rather than the rate of receptor association since the magnitude of the maximal response to the lower affinity analogs was the same as that obtained with cAMP.

It has been proposed that there are several subclasses of surface receptors, a minor set is linked to cGMP accumulation and a major set to activation of adenylate cyclase (Van Haastert and De Wit, 1984). If this is so, our study indicates that the two classes have identical specificity. Furthermore, photoaffinity labeling of intact cells reveals only a single major cAMP binding species, which migrates as a doublet between 40 and 43,000 Da (Theibert *et al.*, 1984) is present on intact cells. A minor amount <5% of the [^{32}P]8-N_3 cAMP is incorporated into a few other bands. Until the specificity of these is determined, we cannot rule out that they could mediate other responses.

Exogenous cAMP profoundly influences the developmental regulation of a host of gene products. The induction of the cAMP receptor itself, adenylate cyclase, and the membrane phosphodiesterase require pulses of low concentrations of cAMP every 6 min from the onset of development (Darmon *et al.*, 1977). Constant low levels of cAMP suppress the expression of these gene products (Klein and Darmon, 1977) and enhance the induction of soluble phosphodiesterase and suppression of its inhibitor (Tsang and Coukell, 1977). Later in development, high concentrations of cAMP induce expression of prestalk specific mRNAs and prevent the rapid degradation of these RNAs when cells are disaggregated (Landfear and Lodish, 1980). The specificity data presented here

provide information necessary for understanding the complex effects of exogenous cAMP on early and late gene expression. This series of analogs can be tested for eliciting expression of a given gene. Comparison of the order of potency with that for the receptor-linked responses and of the major intracellular binding activity should distinguish whether exogenous cAMP acts at surface receptors or intracellularly. Since the results presented here indicate that the receptor-mediated responses have identical specificity, an additional experiment would be required to distinguish whether the rise in intracellular cAMP or another response influences expression of the gene. Caffeine, which prevents activation of adenylate cyclase but no other receptor-mediated responses tested (Brenner and Thoms, 1984), may distinguish this.

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